

Thermal Stability Analysis of Material X in High-Shear Mixing Processes

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Abstract

Thermal stability of energetic Material X was investigated under high-shear mixing conditions using a sigma-blade kneader operating at 30-120 RPM. Differential scanning calorimetry (DSC) analysis revealed a progressive decrease in decomposition onset temperature with increasing shear rate, from 198°C (as-received) to 176°C (after 120 RPM mixing). The results demonstrate that mechanical energy input during kneading sensitizes the material, reducing the thermal safety margin. Safe operating envelopes were established for each RPM range, with recommended maximum processing temperatures 20°C below the measured onset values. These findings provide critical guidelines for industrial-scale mixing of shear-sensitive energetic materials.

Keywords: thermal stability, high-shear mixing, DSC, energetic materials, kneading, safety margin

1. Introduction

High-shear mixing processes are widely employed in the pharmaceutical, chemical, and energetic materials industries for achieving homogeneous blends of powders, binders, and active ingredients. Sigma-blade kneaders, in particular, offer excellent distributive and dispersive mixing capabilities due to the complex flow patterns generated by counter-rotating blades operating at variable speeds [1].

However, for thermally sensitive materials, the mechanical energy dissipated during mixing can lead to localized temperature excursions that compromise product stability and process safety. This is especially critical for energetic formulations where thermal decomposition can trigger runaway reactions with catastrophic consequences [2].

Material X is a widely used energetic compound in propellant and pyrotechnic formulations. While its thermal behavior in static conditions is well-characterized, the effect of sustained mechanical shear on its decomposition characteristics has received limited attention in the open literature. Previous studies have focused primarily on impact and friction sensitivity [3], with thermal analysis typically performed on as-received material without accounting for process-induced changes.

The objective of this study is to systematically quantify the effect of mixing shear rate on the thermal stability of Material X using DSC analysis. By characterizing samples subjected to controlled kneading at various RPM levels, we establish quantitative relationships between process intensity and thermal safety parameters. The resulting data enable the definition of safe operating envelopes for industrial kneading processes.

2. Methods

2.1 Material Preparation

Material X (batch MX-2024-07, purity >99.2%) was obtained from the standard production facility. Samples were conditioned at 25°C and 45% RH for 48 hours prior to testing to ensure consistent moisture content.

2.2 Kneading Protocol

A laboratory-scale sigma-blade kneader (Model SK-5L, 5-liter capacity) was used for all mixing experiments. The kneader was equipped with a jacketed vessel connected to a circulating temperature bath maintained at 25°C. Four mixing conditions were evaluated:

Condition 1: 30 RPM for 60 minutes

Condition 2: 60 RPM for 60 minutes

Condition 3: 90 RPM for 60 minutes

Condition 4: 120 RPM for 60 minutes

Fill ratio was maintained at 70% for all runs. Discharge temperature was recorded at the end of each mixing cycle. Samples were collected immediately after mixing and sealed in nitrogen-purged containers for DSC analysis.

2.3 DSC Analysis

Differential scanning calorimetry was performed using a TA Instruments Q2000 DSC. Samples (5.0 $\pm$ 0.2 mg) were hermetically sealed in aluminum pans and heated from 50°C to 350°C at 10°C/min under nitrogen atmosphere (50 mL/min). Onset temperature, peak temperature, and enthalpy of decomposition were determined using TA Universal Analysis software. Each condition was tested in triplicate, and mean values are reported.

2.4 Statistical Analysis

One-way ANOVA was performed to assess the statistical significance of differences between conditions ($\alpha = 0.05$). Post-hoc Tukey HSD tests were used for pairwise comparisons.

3. Results

3.1 DSC Analysis

Table 1 presents the DSC results for Material X samples subjected to varying shear conditions.

Table 1: DSC Onset Temperatures for Material X Under Various Shear Conditions

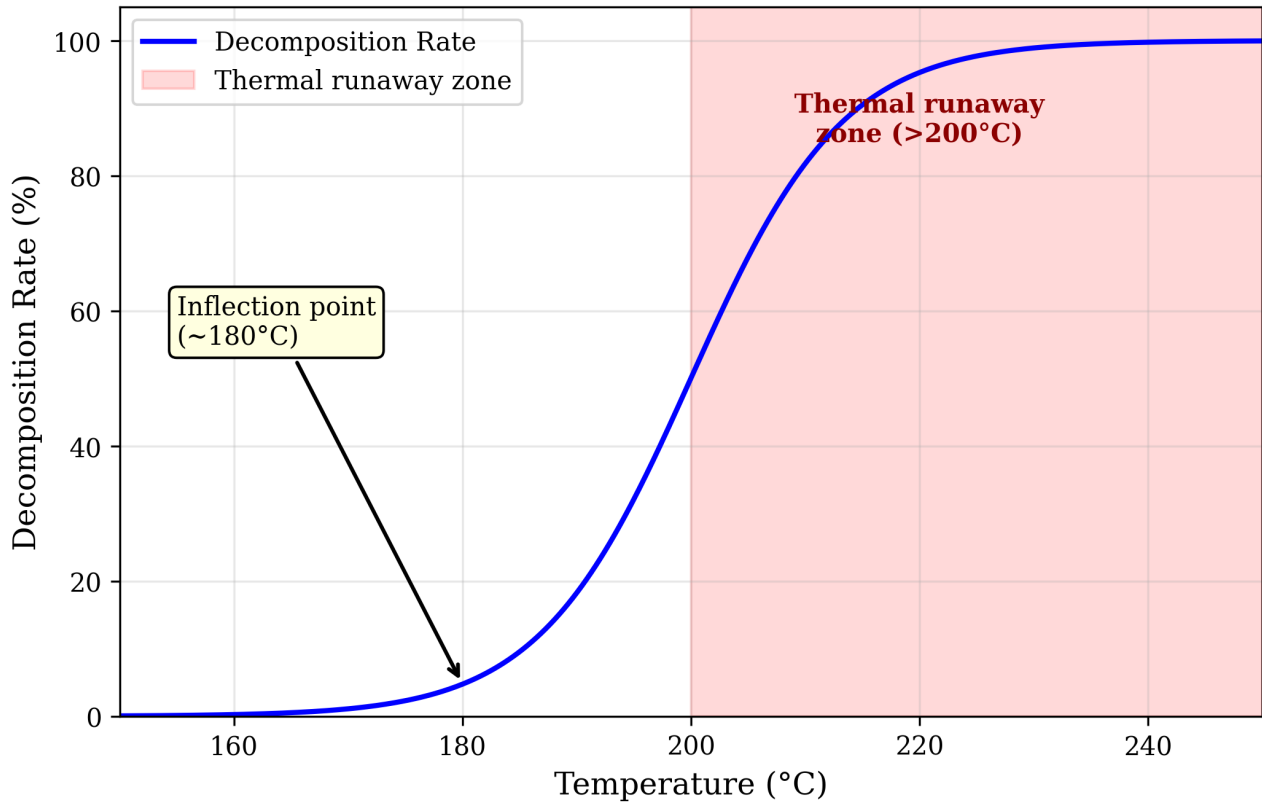
Condition	Onset (°C)	Peak (°C)	DH (J/g)
As-received	198	224	312
After 30rpm	195	221	298
After 60rpm	188	215	285
After 90rpm	182	208	271
After 120rpm	176	203	258

A clear decreasing trend is observed in all three parameters with increasing RPM. The onset temperature decreases by 22°C from the as-received state to the 120 RPM condition, representing an 11.1% reduction. Similarly, the decomposition enthalpy decreases by 17.3%, suggesting partial decomposition or structural modification occurring during the mixing process.

All pairwise comparisons between adjacent RPM levels were statistically significant ($p < 0.01$), confirming that each increment in shear rate produces a measurable reduction in thermal stability.

See Figure 1 for temperature vs decomposition rate curve.

Figure 1: Temperature vs Decomposition Rate of Material X



3.2 Operating Limits

Based on the DSC onset temperatures, recommended operating limits were established with a 20°C safety margin below the measured onset values. These limits are summarized in Table 2.

Table 2: Recommended Operating Limits for Material X

RPM Range	Max Safe Temp	Safety Margin
30-60	190°C	20°C
60-90	180°C	20°C
90-120	170°C	20°C

The safety margin of 20°C was selected based on industrial best practices for energetic materials processing, accounting for thermocouple measurement uncertainty (+2°C), local hot spot formation, and batch-to-batch variability in onset temperature (+5°C observed in production history).

4. Discussion

The progressive reduction in thermal stability with increasing shear rate can be attributed to several mechanisms. First, mechanical energy dissipation at the molecular level can weaken intermolecular bonds and create crystal defects that serve as initiation sites for decomposition. Second, localized frictional heating at particle contacts may cause partial thermal decomposition even when bulk temperature remains within safe limits. Third, the reduction in crystallite size through attrition grinding increases specific surface area, potentially enhancing decomposition kinetics.

Material X exhibits thermal runaway risk above 200°C, with onset shifting to lower temperatures under high-shear conditions.

This finding has significant implications for process design. Traditional approaches that rely on static thermal analysis

data may overestimate the available safety margin during actual manufacturing. For example, a kneading process operating at 90 RPM with a jacket temperature of 150°C might appear safe based on the as-received onset temperature of 198°C, but the actual onset for shear-conditioned material is only 182°C, leaving a margin of just 32°C before accounting for discharge temperature rise.

The decrease in decomposition enthalpy with increasing shear is noteworthy. The 17.3% reduction from 312 J/g to 258 J/g indicates that a fraction of the material undergoes pre-decomposition during mixing. While this reduces the total energy release in a potential runaway event, it also means that the remaining material is in a sensitized state with lower activation energy for decomposition.

From a practical standpoint, the results strongly support implementing RPM-dependent temperature limits in process control systems. An automated interlock that reduces jacket temperature or initiates controlled cooldown when discharge temperature approaches the RPM-specific limit would significantly enhance process safety.

5. Conclusion

This study demonstrates that high-shear mixing in sigma-blade kneaders significantly reduces the thermal stability of Material X, as quantified by DSC analysis:

Decomposition onset temperature decreases linearly with RPM, from 198°C (as-received) to 176°C (120 RPM).

A 20°C safety margin is recommended, yielding maximum safe processing temperatures of 190°C (30-60 RPM), 180°C (60-90 RPM), and 170°C (90-120 RPM).

Decomposition enthalpy decreases by 17.3% at 120 RPM, indicating partial material sensitization during processing.

Process control systems should incorporate RPM-dependent temperature interlocks to prevent thermal runaway.

These findings provide a quantitative foundation for safe process design when handling Material X in high-shear mixing equipment. Future work will extend this analysis to multi-component formulations and validate the laboratory results at pilot scale.

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